🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.AT.9

Graph Inequalities

Lesson 7-5 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can graph the solutions of an inequality on a number line.
Language Objective: I can explain my graph using the words number line, open circle, closed circle, and solution set.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Number line | Open circle | Closed circle | Solution set | Inequality | Boundary point

👆 *Tap any word to see what it means and a picture.*

1. A line with numbers in order used to show values and solutions is a ______________.

2. A hollow circle that shows a value is NOT included, used for < or >, is an ______________.

3. A filled-in circle that shows a value IS included, used for ≤ or ≥, is a ______________.

4. All the values that make an inequality true form the ______________.

5. A math sentence comparing two amounts with <, >, ≤, or ≥ is an ______________.

6 The number where the solution set begins on the number line is the

____________________.

Write About the Math **Guided ESOL writing**

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

When you graphed $x > 4$ and $x \le 7$, why did you use an open circle on one and a closed circle on the other?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Number line** — A line with numbers in order. You use it to show answers.
  Recta numérica — Una línea con números en orden. La usas para mostrar respuestas.
- ☐ **Open circle** — An empty circle showing a number is NOT included.
  Círculo abierto — Un círculo vacío que muestra que un número NO está incluido.
- ☐ **Closed circle** — A filled-in circle showing a number IS included.
  Círculo cerrado — Un círculo relleno que muestra que un número SÍ está incluido.
- ☐ **Solution set** — All the numbers that make the inequality true.
  Conjunto solución — Todos los números que hacen verdadera la desigualdad.
- ☐ **Inequality** — A math sentence comparing two amounts with <, >, ≤, or ≥.
  Desigualdad — Una oración matemática que compara dos cantidades con <, >, ≤ o ≥.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

I used a ____ circle because the symbol ____ means the boundary is ____.

Usé un círculo ____ porque el símbolo ____ significa que el límite ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Because 2:00 itself counts, I use a closed circle at 2.

Evidence: Students choose a closed circle at 2 (because 'no earlier than' includes 2) and shade to the right since later times satisfy $t \geq 2$.

Reasoning: This evidence connects the definition of number line to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **I used a ____ circle because the symbol ____ means the boundary is ____.**
  Usé un círculo ____ porque el símbolo ____ significa que el límite ____.
- **I shaded ____ because the solution set includes ____.**
  Sombreé ____ porque el conjunto solución incluye ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Number line
2. **Notes 2:** Open circle
3. **Notes 3:** Closed circle
4. **Notes 4:** Solution set
5. **Notes 5:** Inequality
6. **Notes 6:** Boundary point
7. **Watch for this mistake:** *A common mistake in Graph Inequalities is treating* $\geq$ *and* $\leq$ *the same as* $>$ *and* $<$ *and drawing the wrong circle. For example, students often graph* $x \geq 6$ *with an OPEN circle (as if it were* $x > 6$*), which wrongly excludes 6 even though 6 itself makes* $x \geq 6$ *true. A second common mistake is choosing the right circle but shading the wrong direction — for* $x < 9$ *students sometimes shade right instead of left. Always check the symbol twice: once for the circle (does it have an 'or equal to' line?) and once for the shading direction (does the symbol point toward larger or smaller numbers?).*
8. **Try It 1:** A. Open circle at 15, shade right — *'Greater than' ($>$) is a strict symbol, so 15 is NOT included — open circle. Larger values work, so shade right.*
9. **Try It 2:** A. Closed circle at 4, shade left — *'Less than or equal to' ($\leq$) includes 4 — closed circle. Smaller values work, so shade left.*